

Lesson 06 Demo 06

Creating Azure SQL DB Geo-Replication

Objective: To create a secondary SQL database in the myResourceGroup database in Azure using the geo-replication

Tools Required: Azure account with administrator access

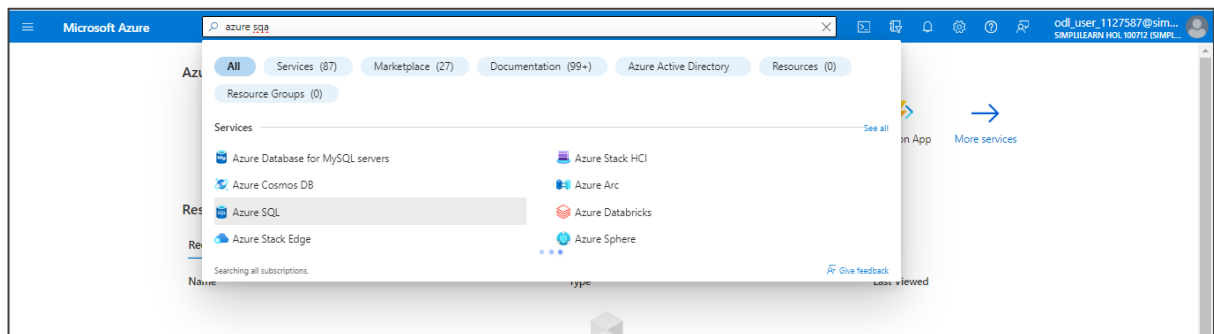
Prerequisites: Azure SQL database

Steps to be followed:

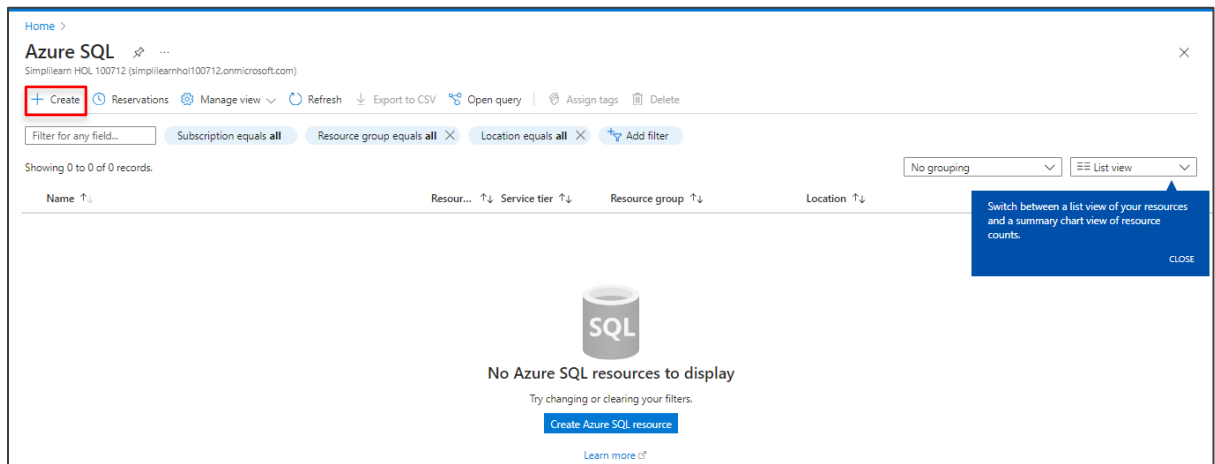
1. Create an Azure SQL database
2. Create a secondary database using geo-replication
3. Perform failover

Step 1: Create an Azure SQL database

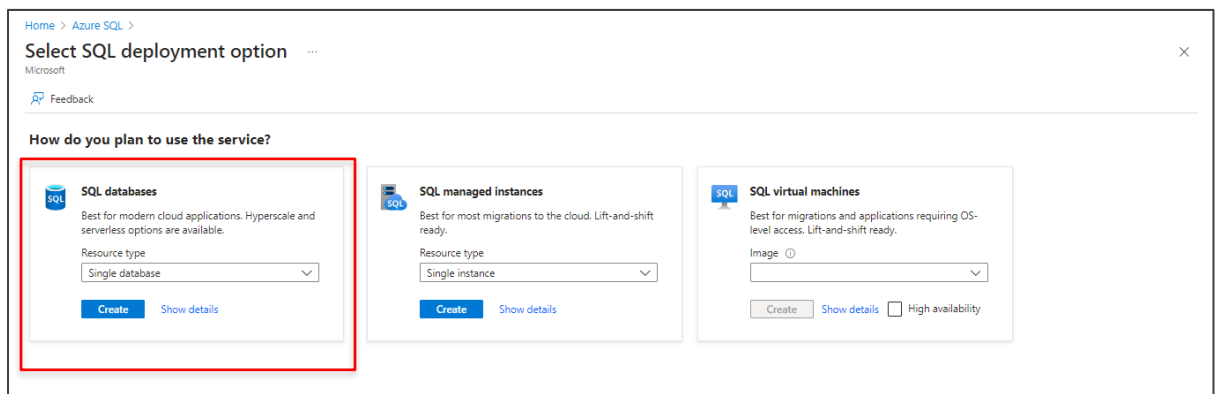
1.1 Log in to the Azure portal and search for **Azure SQL**



1.2 Click on + Create



1.3 Select the SQL databases and click Create



1.4 Under the **Create SQL Database** page, enter the **Resource group** and the **Database name**

Home > Azure SQL > Select SQL deployment option >

Create SQL Database

Microsoft

Basic Networking Security Additional settings Tags Review + create

Create a SQL database with your preferred configurations. Complete the Basics tab then go to Review + Create to provision with smart defaults, or visit each tab to customize. [Learn more](#)

Want to try Azure SQL Database for free? Create a free serverless database with the first 100,000 vCore seconds, 32GB of data, and 32GB of backup storage free per month for the lifetime of the subscription. [Learn more](#)

Apply offer (Preview)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Resource group * [Create new](#)

Database details

Enter required settings for this database, including picking a logical server and configuring the compute and storage resources

Database name *

1.5 Click on **Create new** under **Server** to add a new server

Home > Azure SQL > Select SQL deployment option >

Create SQL Database

Microsoft

Database name *

Server * [Create new](#)

The value must not be empty.

Want to use SQL elastic pool? ☐ Yes ☒ No

Workload environment ☒ Development ☐ Production

Default settings provided for Development workloads. Configurations can be modified as needed.

Compute + storage * [Configure database](#)

Backup storage redundancy

Choose how your PITR and LTR backups are replicated. Geo restore or ability to recover from regional outage is only available when geo-redundant storage is selected.

Backup storage redundancy *

1.6 Enter the **Server name**

Home > Azure SQL > Select SQL deployment option > Create SQL Database >

Create SQL Database Server

Microsoft

Server details

Enter required settings for this server, including providing a name and location. This server will be created in the same subscription and resource group as your database.

Server name * ✓
.database.windows.net

Location * ✓

Authentication

i Azure Active Directory (Azure AD) is now Microsoft Entra ID. [Learn more](#)

Select your preferred authentication methods for accessing this server. Create a server admin login and password to access your server with SQL authentication, select only Microsoft Entra authentication [Learn more](#) or using an existing Microsoft Entra user, group, or application as Microsoft Entra admin [Learn more](#), or select both SQL and Microsoft Entra authentication.

1.7 Select **Use SQL authentication** under the Authentication method, enter the details for **Server admin**, **login Password**, **confirm the password**, and click **OK**.

Home > Azure SQL > Select SQL deployment option > Create SQL Database >

Create SQL Database Server

Microsoft

Authentication

i Azure Active Directory (Azure AD) is now Microsoft Entra ID. [Learn more](#)

Select your preferred authentication methods for accessing this server. Create a server admin login and password to access your server with SQL authentication, select only Microsoft Entra authentication [Learn more](#) or using an existing Microsoft Entra user, group, or application as Microsoft Entra admin [Learn more](#), or select both SQL and Microsoft Entra authentication.

Authentication method

- ☐ Use Microsoft Entra-only authentication
- ☐ Use both SQL and Microsoft Entra authentication
- ☒ Use SQL authentication

Server admin login * ✓

Password * ✓

Confirm password * ✓

OK

1.8 Under the **Basics** tab, create a **Resource group** and enter the **Database name**

The screenshot shows the 'Create SQL Database' wizard in the Azure portal, specifically the 'Basics' tab. The breadcrumb navigation at the top reads 'Home > Azure SQL > Select SQL deployment option >'. The page title is 'Create SQL Database' with a Microsoft logo and a close button. Below the title, there are tabs for 'Basics', 'Networking', 'Security', 'Additional settings', 'Tags', and 'Review + create'. A message box offers a free serverless database trial for 100,000 vCore seconds, 32 GB of data, and 32 GB of backup storage. Below this, the 'Project details' section asks for a subscription (selected as 'Simplilearn HCL 100712') and a resource group (selected as '(New) myRG1'). The 'Database details' section has a 'Database name' field containing 'AZDB'. On the right, a 'Cost summary' sidebar shows pricing for 'General Purpose (GP_S_Gen5_1)' with a cost of 41.6 units. A 'PLEASE CONTACT YOUR RESELLER' banner is at the bottom right.

1.9 Under **Compute + storage**, click on **Configure database**

The screenshot shows the 'Create SQL Database' wizard in the Azure portal, specifically the 'Compute + storage' tab. The breadcrumb navigation at the top reads 'Home > Azure SQL > Select SQL deployment option >'. The page title is 'Create SQL Database' with a Microsoft logo and a close button. Below the title, there are tabs for 'Basics', 'Networking', 'Security', 'Additional settings', 'Tags', and 'Review + create'. The 'Server' dropdown is set to '(new) myazdbserver (East US)'. The 'Want to use SQL elastic pool?' option is set to 'No'. The 'Workload environment' is set to 'Development'. A message box states: 'Default settings provided for Development workloads. Configurations can be modified as needed.' The 'Compute + storage' section is highlighted with an orange box and shows 'General Purpose - Serverless', 'Standard-series (Gen5), 1 vCore, 32 GB storage, zone redundant disabled', and a link to 'Configure database'. The 'Backup storage redundancy' section has three options: 'Locally-redundant backup storage' (selected), 'Zone-redundant backup storage', and 'Geo-redundant backup storage'. At the bottom, there are buttons for 'Review + create' and 'Next: Networking >'. The 'Review + create' button is highlighted in blue.

1.12 Click Create

Home > Azure SQL > Select SQL deployment option >

Create SQL Database

Microsoft

Product details

SQL database by Microsoft

[Terms of use](#) | [Privacy policy](#)

Estimated cost per month

...

Terms

By clicking "Create": (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. For additional details see [Azure Marketplace Terms](#).

Basics

Subscription	Simplilearn HOL 100712
Resource group	myRG1
Region	East US
Database name	AZDB
Server	(new) myazdbserver
Authentication method	SQL authentication
Server admin login	simple
Compute + storage	Basic: 2 GB storage
Backup storage redundancy	Locally-redundant backup storage

Cost summary

Basic (Basic)

Cost per DTU (on -)

DTUs selected: 5

ESTIMATED COST / MONTH

...

PLEASE CONTACT YOUR RESELLER

[Create](#) [Previous](#) [Download a template for automation](#)

The SQL database is created as shown below:

Home >

Microsoft.SqlDatabase.newDatabaseNewServer_66f1779ca62a4fdeab89e | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

Your deployment is complete

Deployment name: Microsoft.SqlDatabase.newDatabaseNewServer_66f... Start time: 10/19/2023, 11:46:11 PM

Subscription: Simplilearn HOL 100712 Correlation ID: bcd95ebb-e360-4d7b-b66d-6637c8f6b4c9

Resource group: myRG1

> Deployment details

< Next steps

[Go to resource](#)

Give feedback

[Tell us about your experience with deployment](#)

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill.

[Set up cost alerts >](#)

Microsoft Defender for Cloud

Secure your apps and infrastructure

[Go to Microsoft Defender for Cloud >](#)

Free Microsoft tutorials

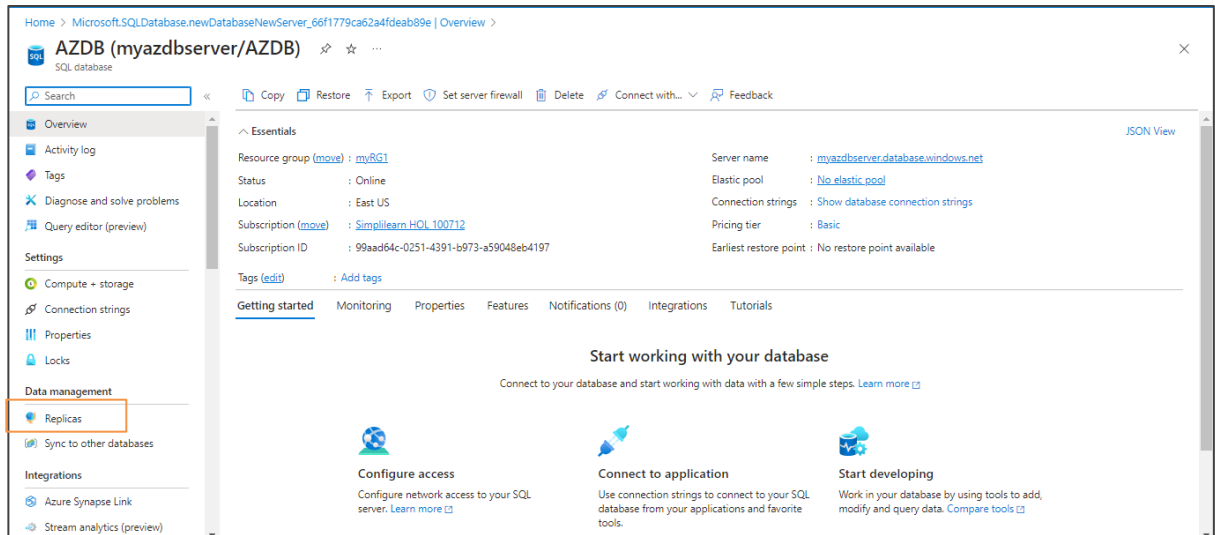
[Start learning today >](#)

Work with an expert

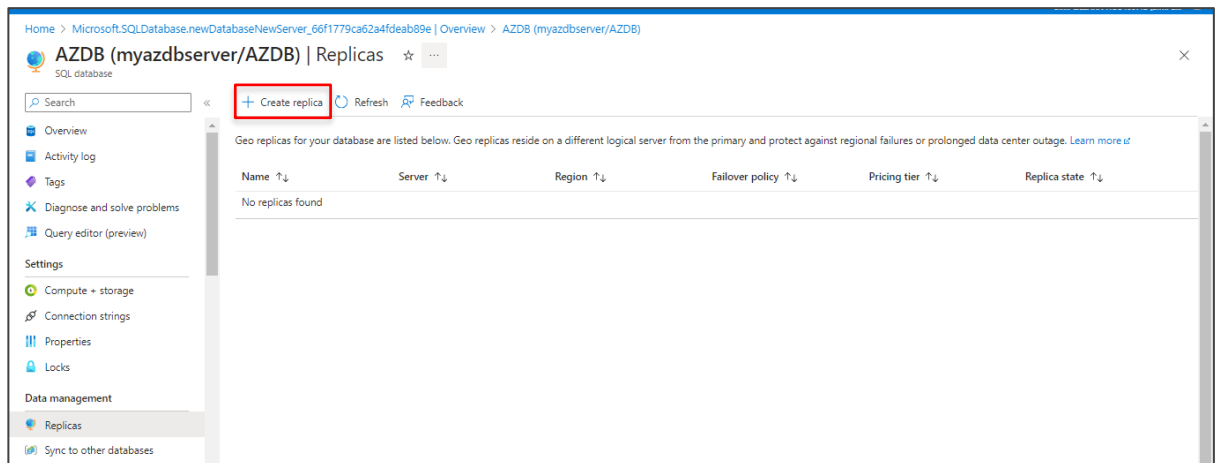
Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.

Step 2: Create a secondary database using geo-replication

2.1 Click on **Go to resource** from the previous step and click on **Replicas** under **Data management**



2.2 Click **+ Create replica**



2.3 In the **Project details**, click on **Create new** to create a **Server** under **Database details**

Home > Microsoft.SQLDatabase.newDatabaseNewServer_66f1779ca62a4fdeab89e | Overview > AZDB (myazdbserver/AZDB) | Replicas >

Create SQL Database - Geo Replica

Microsoft

Basics Review + create

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription

Resource group

Primary database details

Additional settings will be defaulted where possible based on the the primary database.

Primary database

Region

Home > Microsoft.SQLDatabase.newDatabaseNewServer_66f1779ca62a4fdeab89e | Overview > AZDB (myazdbserver/AZDB) | Replicas >

Create SQL Database - Geo Replica

Microsoft

Database details

Enter required settings for this database, including picking a logical server and configuring the compute and storage resources

Database name

Server * [Create new](#)

☒ A database with the same name already exists on this server.

Region

Want to use SQL elastic pool? ☐ Yes ☒ No

Compute + storage * [Configure database](#)

2 GB storage

2.4 Under **Server details**, enter **myserver231** as the **Server name**

Home > Microsoft.SQLDatabase.newDatabaseNewServer_66f1779ca62a4fd5ab89e | Overview > AZDB (myazdbserver/AZDB) | Replicas > Create SQL Database - Geo Replica >

Create SQL Database Server

Microsoft

Server details

Enter required settings for this server, including providing a name and location. This server will be created in the same subscription and resource group as your database.

Server name * ✓
.database.windows.net

Location * ✓

☐ Allow Azure services to access server ⓘ

Authentication

ⓘ Azure Active Directory (Azure AD) is now Microsoft Entra ID. [Learn more](#)

Select your preferred authentication methods for accessing this server. Create a server admin login and password to access your server with SQL authentication, select only Microsoft Entra authentication [Learn more](#) or using an existing Microsoft Entra user, group, or application as Microsoft Entra admin [Learn more](#) or, or select both SQL and Microsoft Entra authentication.

2.5 Under **Authentication method**, choose **Use SQL authentication**, enter **Server admin login**, **Password**, **Confirm password** and Click **OK**

Home > Azure SQL > Select SQL deployment option > Create SQL Database >

Create SQL Database Server

Microsoft

Authentication

ⓘ Azure Active Directory (Azure AD) is now Microsoft Entra ID. [Learn more](#)

Select your preferred authentication methods for accessing this server. Create a server admin login and password to access your server with SQL authentication, select only Microsoft Entra authentication [Learn more](#) or using an existing Microsoft Entra user, group, or application as Microsoft Entra admin [Learn more](#) or, or select both SQL and Microsoft Entra authentication.

Authentication method

☐ Use Microsoft Entra-only authentication
☐ Use both SQL and Microsoft Entra authentication
☒ Use SQL authentication

Server admin login * ✓

Password * ✓

Confirm password * ✓

OK

2.6 Click Review + create

Home > Microsoft.SQLDatabase.newDatabaseNewServer_66f1779ca62a4fdeab89e | Overview > AZDB (myazdbserver/AZDB) | Replicas >

Create SQL Database - Geo Replica

Microsoft

Database details

Enter required settings for this database, including picking a logical server and configuring the compute and storage resources

Database name:

Server * ⓘ: [Create new](#)

Region:

Want to use SQL elastic pool? ⓘ: ☐ Yes ☒ No

Compute + storage * ⓘ: **Basic**
2 GB storage
[Configure database](#)

Transparent data encryption key management

Transparent data encryption encrypts your databases, backups, and logs at rest without any changes to your application. To enable encryption, go to each database. Database level settings if enabled, will override the server level setting. [Learn more](#) ⓘ

Home > Microsoft.SQLDatabase.newDatabaseNewServer_66f1779ca62a4fdeab89e | Overview > AZDB (myazdbserver/AZDB) | Replicas >

Create SQL Database - Geo Replica

Microsoft

[Configure database](#)

Transparent data encryption key management

Transparent data encryption encrypts your databases, backups, and logs at rest without any changes to your application. To enable encryption, go to each database. Database level settings if enabled, will override the server level setting. [Learn more](#) ⓘ

Database level key ⓘ: **Not configured**
[Configure transparent data encryption](#)

Backup storage redundancy

Choose how your PITR and LTR backups are replicated. Geo restore or ability to recover from regional outage is only available when geo-redundant storage is selected.

The default backup storage redundancy setting is taken from the setting of the source.

Backup storage redundancy ⓘ: ☒ Locally-redundant backup storage
☐ Zone-redundant backup storage
☐ Geo-redundant backup storage

[Review + create](#) [Next: Networking >](#)

2.7 Click Create

Home > Microsoft.SQLDatabase.newDatabaseNewServer_66f1779ca62a4fdeab89e | Overview > AZDB (myazdbserver/AZDB) | Replicas >

Create SQL Database - Geo Replica

Microsoft

Basics Networking Additional settings **Review + create**

Product details

SQL database by Microsoft
[Terms of use](#) | [Privacy policy](#)

Estimated cost per month

Terms

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. For additional details see [Azure Marketplace Terms](#).

Basics

Subscription	Simplilearn HOL 100712
Resource group	myRG1
Primary database	AZDB

Home > Microsoft.SQLDatabase.newDatabaseNewServer_66f1779ca62a4fdeab89e | Overview > AZDB (myazdbserver/AZDB) | Replicas >

Create SQL Database - Geo Replica

Microsoft

Region	East US
Database name	AZDB
Server	(new) myserver231
Authentication method	SQL authentication
Server admin login	simple1234
Compute + storage	Basic: 2 GB storage
Database level customer-managed key	Not configured
Database level user assigned managed identity	Not configured
Backup storage redundancy	Locally-redundant backup storage

Networking

Allow Azure services to access server	No
---------------------------------------	----

Additional settings

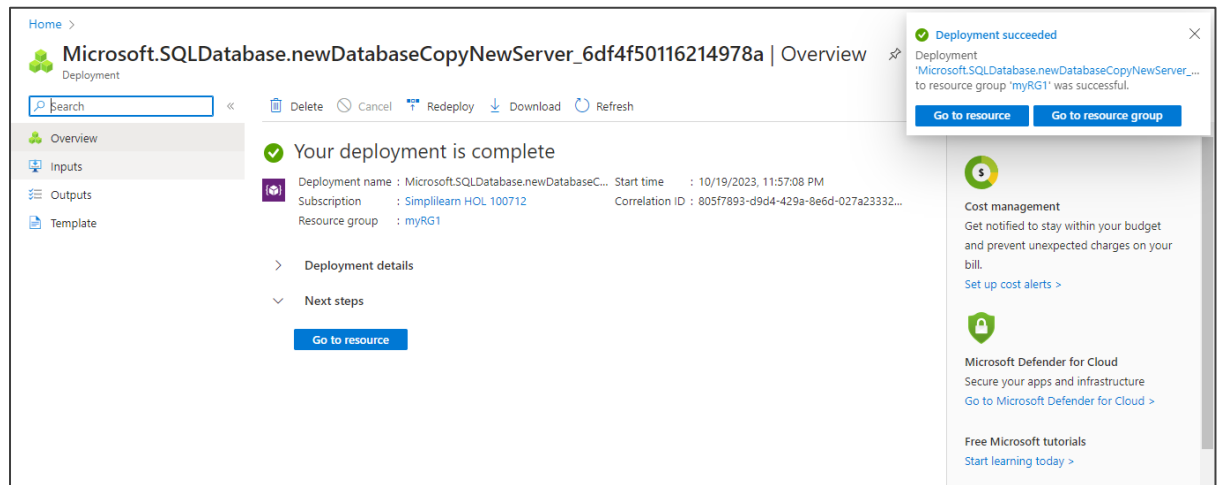
Enable Microsoft Defender for SQL	Not now
-----------------------------------	---------

Create

< Previous

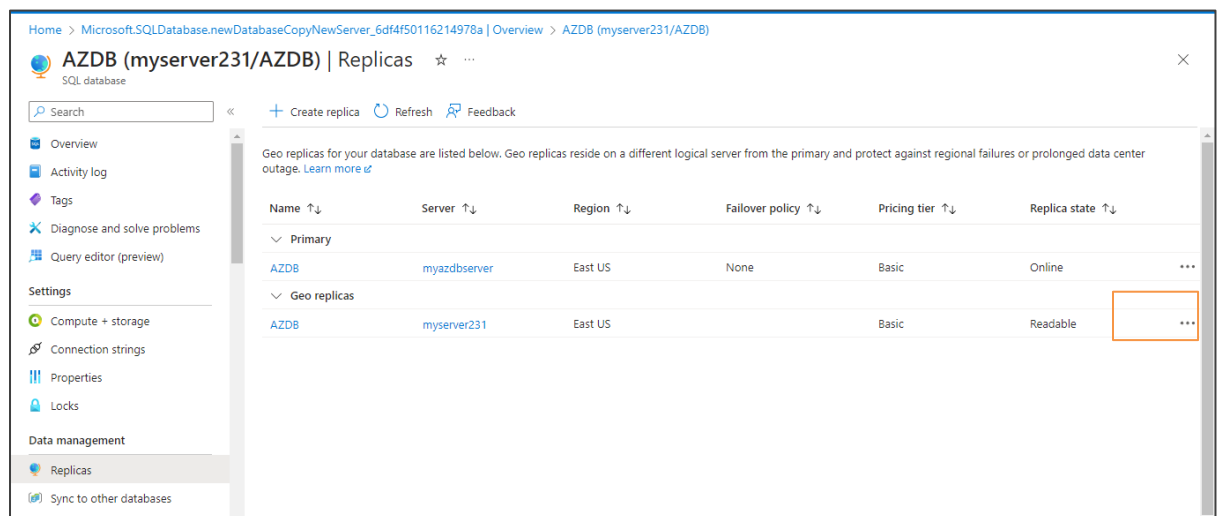
[Download a template for automation](#)

The secondary database is created as shown below:



Step 3: Perform failover

3.1 Click on **Go to resource** in the previous step, select **Replicas** under **Data management**, select the secondary storage (**AZDB**), and click on the three dots



3.2 Click **Forced failover**

Microsoft Azure | Search resources, services, and docs (G+/I)

Home > Microsoft.SQLDatabase.newDatabaseCopyNewServer_6df4f50116214978a | Overview > AZDB (myserver231/AZDB)

AZDB (myserver231/AZDB) | Replicas ☆ ...

SQL database

Search << + Create replica Refresh Feedback

Geo replicas for your database are listed below. Geo replicas reside on a different logical server from the primary and protect against regional failures or prolonged data center outage. [Learn more](#)

Name ↑↓	Server ↑↓	Region ↑↓	Failover policy ↑↓	Pricing tier ↑↓	Replica state ↑↓	
▼ Primary						
AZDB	myazdbserver	East US	None	Basic	Online	...
▼ Geo replicas						
AZDB	myserver231	East US		Basic	Readonly	...

- Pin to dashboard
- Planned failover
- Forced failover**
- Stop replication

3.3 Click **Yes**

Microsoft Azure | Search resources, services, and docs (G+/I)

Home > Microsoft.SQLDatabase.newDatabaseCopyNewServer_6df4f50116214978a | Overview > AZDB (myserver231/AZDB)

AZDB (myserver231/AZDB) | Replicas ☆ ...

SQL database

Search << + Create replica Refresh Feedback

Geo replicas for your database are listed below. Geo replicas reside on a different logical server from the primary and protect against regional failures or prolonged data center outage. [Learn more](#)

Name ↑↓	Server ↑↓	Region ↑↓	Failover policy ↑↓	Pricing tier ↑↓	Replica state ↑↓	
▼ Primary						
AZDB	myazdbserver	East US	None	Basic	Online	...
▼ Geo replicas						
AZDB	myserver231	East US		Basic	Readonly	...

Forced failover

Forced failover will switch myserver231/AZDB to the primary role. The old primary myazdbserver/AZDB will automatically become the new secondary if it is online. All TDS sessions will be disconnected. This operation may cause data loss. Are you sure you want to proceed?

Yes **No**

3.4 Click **Refresh** to view the **secondary database**

Microsoft Azure

Home > Microsoft.SQLDatabase.newDatabaseCopyNewServer_6df4f50116214978a | Overview > AZDB (myserver231/AZDB)

AZDB (myserver231/AZDB) | Replicas

SQL database

Search resources, services, and docs (G+)

odl_user_1127587@sim... SIMPLILEARN HOL 100712 (SIMPLILEARN)

Request submitted for failover replication
Request submitted for failover replication to secondary server: myserver231, secondary database: AZDB

Geo replicas for your database are listed below. Geo replicas reside on a different logical server from the primary and protect against regional failures or prolonged data center outage. [Learn more](#)

Name ↑↓	Server ↑↓	Region ↑↓	Fallover policy ↑↓	Pricing tier ↑↓	Replica state ↑↓
▼ Primary					
AZDB	myazdbserver	East US	None	Basic	Online
▼ Geo replicas					
AZDB	myserver231	East US		Basic	Readable

The final output is as shown below:

Microsoft Azure

Home > Microsoft.SQLDatabase.newDatabaseCopyNewServer_6df4f50116214978a | Overview > AZDB (myserver231/AZDB)

AZDB (myserver231/AZDB) | Replicas

SQL database

Search resources, services, and docs (G+)

odl_user_1127587@sim... SIMPLILEARN HOL 100712 (SIMPLILEARN)

Request submitted for failover replication
Request submitted for failover replication to secondary server: myserver231, secondary database: AZDB

Geo replicas for your database are listed below. Geo replicas reside on a different logical server from the primary and protect against regional failures or prolonged data center outage. [Learn more](#)

Name ↑↓	Server ↑↓	Region ↑↓	Fallover policy ↑↓	Pricing tier ↑↓	Replica state ↑↓
▼ Primary					
AZDB	myserver231	East US	None	Basic	Pending
▼ Geo replicas					
AZDB	myazdbserver	East US		Basic	Failover...

By following the above steps, you have successfully created a secondary SQL database in the myResourceGroup database in Azure using geo-replication.